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IMAGE FORMING APPARATUS

Abstract

An image forming apparatus includes a casing, an image carrier unit, and a waste toner container. The image carrier unit is housed in the casing. The image carrier unit includes an image carrier on which an electrostatic latent image is formed. The waste toner container is housed in the casing. The waste toner container collects waste toner discharged from the image carrier unit. The image carrier unit is insertable or removable from one side in a first direction in the casing. The waste toner container is insertable or removable from another side of the first direction in the casing.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION

[0001] This application claims the benefit of Japanese Priority Patent Application JP 2024-029735 filed Feb. 29, 2024, under 35 U.S.C. 119, the entire contents of which are incorporated herein by reference.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates to an image forming apparatus.

BACKGROUND OF THE DISCLOSURE

[0003] Japanese Patent Application Laid-open No. 2017-68211 has disclosed an image forming apparatus. The image forming apparatus according to Japanese Patent Application Laid-open No. 2017-68211 develops electrostatic latent images formed on photosensitive drums serving as image carriers with toner as toner images. The toner images on the photosensitive drums are transferred to a transferring belt. Then, the toner images on the transferring belt are further transferred to a paper sheet. The image forming apparatus according to Japanese Patent Application Laid-open No. 2017-68211 has an opening on its side surface. The opening is used for mounting and removing a toner cartridge. The toner cartridge is arranged in the opening for mounting and removing the toner cartridge. The toner cartridge includes a supply toner storage part and a waste toner collection part. The waste toner collection part collects waste toner discharged from drum units including the image carriers.

SUMMARY OF THE DISCLOSURE

[0004] An image forming apparatus according to an aspect of the present disclosure includes a casing, an image carrier unit, and a waste toner container. The image carrier unit is housed in the casing. The image carrier unit includes an image carrier on which an electrostatic latent image is formed. The waste toner container collects waste toner discharged from the image carrier unit. The image carrier unit is insertable or removable from one side in a first direction in the casing. The waste toner container is insertable or removable from another side of the first direction in the casing.

[0005] These and other objects, features and advantages of the present disclosure will become more apparent in light of the following detailed description of best mode embodiments thereof, as illustrated in the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a schematic perspective view of an image forming apparatus according to an embodiment of the present disclosure.

[0007] FIG. 2 is a schematic cross-sectional view of the image forming apparatus.

[0008] FIG. 3 is a schematic side view showing a state in which an image carrier unit and a waste toner container are inserted and removed with respect to a casing of the image forming apparatus.

[0009] FIG. 4 is a schematic perspective view of the image carrier unit.

[0010] FIG. 5 is a schematic perspective view of the waste toner container.

[0011] FIG. 6 is a schematic cross-sectional view of a discharge passageway and a reception passageway.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

[0012] Hereinafter, an embodiment of an image forming apparatus **10** according to the present disclosure will be described with reference to the drawings. Note that in the figures, the same or corresponding portions will be denoted by the same reference signs and the descriptions will not be

repeated.

[0013] First of all, the entire configuration of the image forming apparatus **10** according to the embodiment of the present disclosure will be described with reference to FIG. **1**. FIG. **1** is a schematic perspective view of the image forming apparatus **10**. The image forming apparatus **10** is, for example, a multifunction peripheral with printer, scanner, and copier functions. The image forming apparatus **10** may have a facsimile function. The image forming apparatus **10** forms an image on a paper sheet. In this embodiment, the image forming apparatus **10** forms an image on a paper sheet by an electrophotographic method.

[0014] As shown in FIG. **1**, in this embodiment, the image forming apparatus **10** includes a casing **11**, a paper delivery tray **15**, an image scanning part **16**, and an operation part **17**. An image is formed on a paper sheet inside the casing **11**. Then, the paper sheet with the image formed thereon is delivered to the paper delivery tray **15**.

[0015] The image scanning part **16** scans a document not shown in FIG. **1** and generates image data. The image forming apparatus **10** advantageously forms an image on a paper sheet on the basis of the image data generated by the image scanning part **16**, for example. Alternatively, the image forming apparatus **10** may form an image on a paper sheet on the basis of image data sent to the image forming apparatus **10** from an external apparatus (e.g., a computer).

[0016] The operation part **17** includes a display part and an input part. The display part is, for example, a liquid-crystal display and notifies a user of necessary information. The input part is, for example, a keypad and receives an instruction from the user. The operation part **17** may be a touch panel display where the display part and the input part are integrated.

[0017] The casing **11** houses mechanical components and the like for forming an image on a paper sheet. A rear cover **12** is provided on a surface at one side AB of a first direction A in the casing **11**. Moreover, a front cover **13** is provided on a surface at another side AF of the first direction A in the casing **11**. The first direction A is a direction that intersects with a gravity direction G (vertical direction) in the casing **11**.

[0018] Hereinafter, in this specification, the first direction A will be sometimes referred to as a front-back direction A, the one side AB of the first direction A (front-back direction A) will be sometimes referred to as back AB, and the other side AF of the first direction A will be sometimes referred to as front AF. The rear cover **12** opens and closes the surface in the back AB of the casing **11**. The front cover **13** opens and closes a surface of the casing **11** in the front AF. Moreover, a paper feeding cassette **14** that stores paper sheets is provided below the front cover **13** at the front AF of the casing **11**.

[0019] Next, an image forming operation of the image forming apparatus **10** will be described with reference to FIG. **2**. FIG. **2** is a schematic cross-sectional view of the image forming apparatus **10**. FIG. **2** shows a cross-sectional view as the casing **11** is taken along a vertical surface in an orientation parallel to the first direction A. In FIG. **2**, a left-right direction is the first direction A and an up-down direction is the gravity direction G.

[0020] The casing **11** houses the image carrier unit **20** and the waste toner container **30**. Moreover, a toner feeding container **38**, a developing unit **42**, a transferring belt **45**, and a fixation apparatus **48** are further arranged in the casing **11**. Although not shown in FIG. **2**, an exposure device including a laser light emitter is arranged in the casing **11**. The transferring belt **45** is arranged along a circulation path extending in the gravity direction G inside the casing **11**. The transferring belt **45** is suspended over a plurality of rollers. By rotating the plurality of rollers, the transferring belt **45** is driven to circulate on the circulation path. Moreover, a conveyance path **40** of paper sheets S, which are conveyed from the paper feeding cassette **14** to the paper delivery tray **15** through a part of the circulation path of the transferring belt **45**, is provided inside the casing **11**.

[0021] The image carrier unit **20** includes an image carrier **21**, a discharge passageway **22**, a charging device **23**, and a cleaner **24**. An electrostatic latent image is formed on the image carrier **21**. The discharge passageway **22** guides waste toner. The charging device **23** charges the surface of

the image carrier **21**. The cleaner **24** removes toner remaining on the image carrier **21** as waste toner. The image carrier unit **20** further includes an exit port **25** and an exit port shutter **26**. The exit port **25** is used for discharging the waste toner. The exit port shutter **26** opens and closes the exit port **25**. In FIG. 2, the discharge passageway **22** extends in the first direction A and guides the waste toner to the exit port **25** to the other side AF (front AF) of the first direction A. Moreover, the exit port **25** is formed on the discharge passageway **22**.

[0022] The waste toner container **30** is a container that collects the waste toner discharged from the image carrier unit **20**. The waste toner container **30** includes a reception passageway **32**, a reception inlet **35**, and a reception inlet shutter **36**. The reception passageway **32** guides the waste toner. The reception inlet **35** receives the waste toner discharged from the exit port **25**. The reception inlet shutter **36** opens and closes the reception inlet **35**. In FIG. 2, the reception passageway **32** extends in the first direction A and guides the waste toner received from the reception inlet **35** to the other side AF (front AF) of the first direction A. Moreover, the reception inlet **35** is formed on the reception passageway **32**.

[0023] The toner feeding container **38** is a container that supplies toner to be used in the image carrier unit **20**. The toner feeding container **38** includes a supply passageway **39** that extends in the first direction A. The toner feeding container **38** guides toner to the one side AB (back AB) of the first direction A through the supply passageway **39** and supplies the toner to the developing unit **42**. The toner supplied to the developing unit **42** is used in the image carrier unit **20**.

[0024] In FIG. 2, the toner feeding container **38** is a container that is integrated with the waste toner container **30**. Since the toner feeding container **38** and the waste toner container **30** are integrated, only a small space in the casing **11** is required for supplying toner and collecting waste toner. Therefore, the image forming apparatus **10** can be downsized. Moreover, the toner feeding container **38** and the waste toner container **30** can be replaced at a time. It improves the operability of the image forming apparatus **10**. Note that the toner feeding container **38** is not limited to the one integrated with the waste toner container **30**, and the toner feeding container **38** may be a container separated from the waste toner container **30**. In a case where the toner feeding container **38** is a container separated from the waste toner container **30**, the user can replace only one of the toner feeding container **38** and the waste toner container **30** that the user should replace.

[0025] The image carrier **21** of the image carrier unit **20** is a member on which an electrostatic latent image is formed. The image carrier **21** of the image carrier unit **20** is, for example, a photosensitive drum. The image carrier **21** is arranged to rotate at a position close to the transferring belt **45**.

[0026] The charging device **23** charges the surface of the image carrier **21**. The exposure device not shown in FIG. 2 emits laser light to the image carrier **21** and forms an electrostatic latent image based on the image data on the surface of the image carrier **21**. The developing unit **42** develops the electrostatic latent image of the image carrier **21** with toner. A toner image is thus formed on the surface of the image carrier **21**. The toner image on the image carrier **21** is transferred to the transferring belt **45**. Note that the cleaner **24** removes toner remaining on the image carrier **21** as waste toner.

[0027] The paper feeding cassette **14** stores the plurality of paper sheets S. The paper sheets S in the paper feeding cassette **14** are fed to the conveyance path **40** one by one. The transferring belt **45** transfers a toner image to the paper sheet S transported to the conveyance path **40**. The toner image transferred to the paper sheet S is fixed to the paper sheet S by the fixation apparatus **48**. Then, the paper sheet S with the toner image fixed thereto is delivered to the paper delivery tray **15**. The image forming operation of the image forming apparatus **10** is performed in the above-mentioned manner. Note that the paper sheet S may be transported to the conveyance path **40** from the back AB of the casing **11** with the rear cover **12** of the casing **11** opened.

[0028] Next, insertion and removal of the image carrier unit **20** and the waste toner container **30** with respect to the casing **11** will be described with reference to FIGS. 2 and 3. FIG. 3 is a

schematic side view showing a state in which the image carrier unit **20** and the waste toner container **30** are inserted and removed with respect to the casing **11** of the image forming apparatus **10**.

[0029] As shown in FIG. **3**, the image carrier unit **20** is insertable or removable from the one side AB (back AB) of the first direction A in the casing **11**. The waste toner container **30** is insertable or removable from the other side AF (front AF) of the first direction A in the casing **11**. Specifically, in a case where the rear cover **12** provided on the surface at the one side AB of the first direction A in the casing **11** is opened, the image carrier unit **20** is insertable or removable with respect to the casing **11** from the one side AB of the first direction A. Moreover, in a case where the front cover **13** provided on the surface at the other side AF of the first direction A in the casing **11** is opened, the waste toner container **30** is insertable or removable with respect to the casing **11** from the other side AF of the first direction A. Note that since the waste toner container **30** is a container integrated with the toner feeding container **38** in this embodiment, the waste toner container **30** and the toner feeding container **38** are insertable or removable with respect to the casing **11** together from the other side AF of the first direction A.

[0030] In the image forming apparatus **10** according to this embodiment, the insertion and removal of the image carrier unit **20** with respect to the casing **11** from the one side AB of the first direction A and the insertion and removal of the waste toner container **30** with respect to the casing **11** from the other side AF of the first direction A can be independently performed. Therefore, in accordance with the image forming apparatus **10** according to this embodiment, the image carrier unit **20** and the waste toner container **30** are independently replaceable. That is, the user can perform the replacement of the image carrier unit **20** and the replacement of the waste toner container **30** at the same time. It improves the operability of the image forming apparatus **10**.

[0031] By the way, regarding the image forming apparatus according to Japanese Patent Application Laid-open No. 2017-68211, a drum unit including an image carrier is arranged more inside than a toner cartridge including a waste toner collection part inside the image forming apparatus. In the image forming apparatus according to Japanese Patent Application Laid-open No. 2017-68211, it is necessary to first remove the outside toner cartridge from the image forming apparatus for removing the inside drum unit from the image forming apparatus. Therefore, even in a case where only the drum unit should be replaced, it is necessary to first remove the toner cartridge from the image forming apparatus. In a case where both the drum unit and the toner cartridge should be replaced, the toner cartridge is replaced after the drum unit is replaced. Therefore, in the image forming apparatus according to Japanese Patent Application Laid-open No. 2017-68211, the user cannot perform the drum unit replacement and the toner cartridge replacement at the same time. In contrast, in accordance with the image forming apparatus **10** according to this embodiment, the unit including the image carrier and the waste toner container are independently replaceable.

[0032] The image carrier unit **20** and the waste toner container **30** are inserted into the casing **11** from the back AB and the front AF of the casing **11**, respectively, and are housed in the casing **11**. In a state in which the image carrier unit **20** and the waste toner container **30** are housed in the casing **11**, as shown in FIG. **2**, the image carrier unit **20** is disposed at a position closer to the one side AB of the first direction A than the waste toner container **30** inside the casing **11**. That is, the image carrier unit **20**, which is inserted and removed with respect to the casing **11** from the back AB, is disposed at a position closer to the back AB inside the casing **11**. The waste toner container **30**, which is inserted and removed with respect to the casing **11** from the front AF, is disposed at a position closer to the front AF inside the casing **11**.

[0033] Therefore, in the image forming apparatus **10** according to this embodiment, the user can remove only a part to be replaced from the casing **11** when only one of the image carrier unit **20** and the waste toner container **30** should be replaced.

[0034] Note that a mechanism that guides the insertion and removal of the image carrier unit **20**

and the waste toner container **30** is favorably provided so that the image carrier unit **20** and the waste toner container **30** can be disposed at appropriate positions inside the casing **11**. For example, a rail that guides the image carrier unit **20** and a rail that guides the waste toner container **30** are advantageously provided on the inner surface of the casing **11**.

[0035] Next, the discharge passageway **22** and the exit port **25** of the image carrier unit **20** and the reception passageway **32** and the reception inlet **35** of the waste toner container **30** will be described with reference to FIGS. **2**, **4**, **5**, and **6**. FIG. **4** is a schematic perspective view of the image carrier unit **20**. FIG. **5** is a schematic perspective view of the waste toner container **30**. FIG. **6** is a schematic cross-sectional view of the discharge passageway **22** and the reception passageway **32**.

[0036] The image carrier unit **20** in FIG. **4** includes a thin and long image carrier accommodation unit **28** and an image carrier drive unit **29**. The image carrier accommodation unit **28** extends in a second direction W. The image carrier drive unit **29** is provided at an end portion of the image carrier accommodation unit **28**. Note that the second direction W is a direction that intersects with both the gravity direction G and the first direction A and corresponds to a width direction of the paper sheet S.

[0037] The image carrier accommodation unit **28** houses image carrier **21** (FIG. **2**). Since the toner image transferred to the entire paper sheet S in the width direction is formed on the surface of the image carrier **21**, the image carrier **21** has a width dimension (dimension in the second direction W) larger than the width dimension of the paper sheet S. Therefore, as shown in FIG. **4**, the image carrier accommodation unit **28** has a thin and long shape extending in the second direction W. The image carrier drive unit **29** rotates the image carrier **21** in the image forming operation of the image forming apparatus **10**.

[0038] The discharge passageway **22** extends to the other side AF (front AF) of the first direction A from the image carrier drive unit **29**. The waste toner removed from the surface of the image carrier **21** is collected to the image carrier drive unit **29** from the image carrier accommodation unit **28**. Then, the waste toner is guided to the exit port **25** through the discharge passageway **22**.

[0039] The exit port **25** is provided on a lower surface of the discharge passageway **22** in the gravity direction G, in vicinity of an end portion of the discharge passageway **22**, which is located at the front AF. The exit port shutter **26** is provided in a lower position than the exit port **25**. The exit port shutter **26** opens and closes the exit port **25**. Specifically, for example, the exit port shutter **26** advantageously opens and closes the exit port **25** by sliding in the first direction A along the lower surface of the discharge passageway **22**. Moreover, the exit port shutter **26** is advantageously biased in a direction to close the exit port **25** through an elastic member (e.g., spring) (not shown).

[0040] Next, the waste toner container **30** in FIG. **5** has an elongated shape extending in the first direction A. Note that the waste toner container **30** in FIG. **5** is an integrated-type cartridge **33** that is integrated with the toner feeding container **38**. In the integrated-type cartridge **33** in FIG. **5**, the toner feeding container **38** has a shape to cover an upper surface and a side surface of the waste toner container **30**. Specifically, the toner feeding container **38** has a shape to cover an upper surface of the waste toner container **30** in the gravity direction G and extend downwards in the gravity direction G from an end portion in the second direction W. In FIG. **5**, the toner feeding container **38** has an inverted L-shape as viewed in the first direction A.

[0041] The reception passageway **32** of the waste toner container **30** extends to the back AB from an end surface of the waste toner container **30**, which is located at the one side AB (back AB) of the first direction A. Moreover, the supply passageway **39** of the toner feeding container **38** extends to the back AB from the end surface of the toner feeding container **38** in the back AB.

[0042] The reception inlet **35** of the waste toner container **30** is provided on an upper surface of the reception passageway **32** in the gravity direction G, in vicinity of an end portion of the reception passageway **32**, which is located at the back AB. The reception inlet shutter **36** is provided in an upper position than the reception inlet **35**. The reception inlet shutter **36** opens and closes the

reception inlet **35**. Specifically, for example, the reception inlet shutter **36** advantageously opens and closes the reception inlet **35** by sliding in the first direction A along the upper surface of the reception passageway **32**. Moreover, the reception inlet shutter **36** is advantageously biased in a direction to close the reception inlet **35** through an elastic member (e.g., spring) (not shown).

[0043] As shown in FIG. 2, when the image carrier unit **20** and the waste toner container **30** are arranged inside the casing **11**, the discharge passageway **22** of the image carrier unit **20** is arranged in an upper position in the gravity direction G than the reception passageway **32** of the waste toner container **30**.

[0044] Portions of the image carrier unit **20** and the waste toner container **30**, excluding the discharge passageway **22** and the reception passageway **32**, can be disposed at positions that are distant from each other in the first direction A in the casing **11** because of the discharge passageway **22** and the reception passageway **32** that guide the waste toner in the first direction A.

[0045] For example, the image carrier accommodation unit **28** and the image carrier drive unit **29**, and portions occupying a large space, such as a waste toner storage portion of the waste toner container **30**, can be arranged in regions distant from each other. The waste toner is guided to the waste toner container **30** from the image carrier unit **20** even when the image carrier unit **20** and the waste toner container **30** are distant from each other because of the discharge passageway **22** and the reception passageway **32**. Therefore, the degree of freedom in the lay-out of components of the image forming apparatus **10** increases. Moreover, arranging the discharge passageway **22** in an upper position than the reception passageway **32** enables the discharge passageway **22** and the reception passageway **32** to be arranged in partially overlapping regions in a planar view. Downsizing of the image forming apparatus **10** can be thus achieved.

[0046] When the image carrier unit **20** and the waste toner container **30** are arranged inside the casing **11**, as shown in FIG. 6, the exit port **25** and the reception inlet **35** are disposed at positions to face each other (in FIG. 6, positions to face each other in up and down directions in the gravity direction G).

[0047] The exit port shutter **26** and the reception inlet shutter **36** favorably close the exit port **25** and the reception inlet **35**, respectively, through the elastic members or the like when the exit port **25** and the reception inlet **35** do not face each other. The exit port shutter **26** opens the exit port **25** when the exit port **25** and the reception inlet **35** are disposed at positions to face each other. The reception inlet shutter **36** opens the reception inlet **35**. Therefore, in accordance with the image forming apparatus **10** according to this embodiment, the exit port **25** and the reception inlet **35** are opened when the waste toner is reliably transported from the image carrier unit **20** to the waste toner container **30**. Therefore, the waste toner is prevented from being scattered from the exit port **25** or the reception inlet **35**.

[0048] Specifically, the exit port **25** and the reception inlet **35** are opened by the exit port shutter **26** and the reception inlet shutter **36** sliding each other. When the image carrier unit **20** is inserted into the casing **11** from the one side AB of the first direction A and the waste toner container **30** is inserted into the casing **11** from the other side AF of the first direction A, the discharge passageway **22** and the reception passageway **32** approach each other in the first direction A.

[0049] The discharge passageway **22** is arranged in an upper position in the gravity direction G than the reception passageway **32**. The discharge passageway **22** is disposed at a position where the exit port shutter **26** sliding along the lower surface of the discharge passageway **22** is brought into contact with the reception inlet shutter **36** sliding along the upper surface of the reception passageway **32** specifically when the discharge passageway **22** and the reception passageway **32** approach each other. Therefore, when the image carrier unit **20** and the waste toner container **30** are inserted into the casing **11**, the exit port shutter **26** and the reception inlet shutter **36** are brought into contact with each other to slide each other in the first direction A, and the exit port **25** and the reception inlet **35** are opened as shown in FIG. 6. The exit port shutter **26** and the reception inlet shutter **36** favorably have a shape with which they hook each other when they are brought into

contact with each other in the first direction A to slide each other.

[0050] Note that a mechanism that positions the image carrier unit **20** and the waste toner container **30** is favorably provided so that the exit port **25** and the reception inlet **35** can be reliably opened when the image carrier unit **20** and the waste toner container **30** are arranged inside the casing **11**. For example, a rail that guides the image carrier unit **20** and the waste toner container **30** and a stopper that positions the image carrier unit **20** and the waste toner container **30** are advantageously provided on the inner surface of the casing **11**.

[0051] As shown in FIG. **6**, a waste toner discharge screw **27** that extends in the first direction A is provided inside the discharge passageway **22**. Moreover, a waste toner reception screw **37** that extends in the first direction A is provided inside the reception passageway **32**. The waste toner discharge screw **27** rotates about the first direction A by receiving driving force from the image carrier drive unit **29** and the like and conveys the waste toner in the image carrier unit **20** to the exit port **25** to the other side AF of the first direction A.

[0052] The waste toner conveyed to the exit port **25** falls from the exit port **25** due to the gravity, passes through the reception inlet **35** facing the exit port **25**, and is received in the reception passageway **32**. The waste toner reception screw **37** in the reception passageway **32** rotates about the first direction A and conveys the waste toner received from the reception inlet **35** to the waste toner container **30** to the other side AF of the first direction A. In the above-mentioned manner, the waste toner discharged from the image carrier unit **20** is collected to the waste toner container **30**.

[0053] Although the single image carrier unit **20** and the single waste toner container **30** are described above for the description, a plurality of image carrier units **20** and a plurality of waste toner containers **30** may be provided. For example, in a case where the image forming apparatus **10** is a color printer using toners in four colors, four image carrier units **20** and four waste toner containers **30** are advantageously provided, corresponding to the toners in four colors, respectively. Note that the toners in four colors are advantageously, for example, four colors of YMCK (Y: yellow, M: magenta, C: cyan, K: black). Moreover, a plurality of developing units **42** is advantageously also provided, corresponding to the respective colors for the image carrier units **20** and the waste toner containers **30**.

[0054] The plurality of image carrier units **20** is advantageously insertable or removable from the one side AB of the first direction A at different heights with respect to the casing **11**. The plurality of image carrier units **20** is arranged in a height direction (gravity direction G) with respect to the transferring belt **45** inside the casing **11**.

[0055] The plurality of image carrier units **20** advantageously transfers toner images in the corresponding colors to the transferring belt **45**. The plurality of image carrier units **20** transfers the toner images in the corresponding colors to the transferring belt **45**. In this manner, a multiple-color (e.g., full-color) toner image are formed on the transferring belt **45**. Then, multiple-color image formation on the paper sheet S is performed by transferring the multiple-color toner image to the paper sheet S.

[0056] The plurality of waste toner containers **30** is advantageously insertable or removable from the other side AF of the first direction A at different heights with respect to the casing **11**. The plurality of waste toner containers **30** collects the waste toners discharged from the corresponding image carrier units **20** inside the casing **11**.

[0057] Hereinabove, the embodiment of the present disclosure has been described with reference to the drawings. Note that the present disclosure is not limited to the above-mentioned embodiment, and may be carried out in various aspects without departing from the gist of the present disclosure. For easy understanding, the drawings each schematically show configurations of elements mainly, and the thickness, the length, the number of items, the intervals, and the like of each component shown in the figure are different from the actual ones for the sake of convenience for creating the drawings. Moreover, the material, the shape, the dimensions, and the like of each component shown in the above-mentioned embodiment, and there are no particular limitations and various

modifications can be made without substantially departing from the configurations of the present disclosure.

[0058] It should be understood by those skilled in the art that various modifications, combinations, sub-combinations and alterations may occur depending on design requirements and other factors insofar as they are within the scope of the appended claims or the equivalents thereof.

Claims

1. An image forming apparatus, comprising: a casing; an image carrier unit that is housed in the casing and includes an image carrier on which an electrostatic latent image is formed; and a waste toner container that is housed in the casing and collects waste toner discharged from the image carrier unit, wherein the image carrier unit is insertable or removable from one side of a first direction in the casing, and the waste toner container is insertable or removable from another side of the first direction in the casing.
 2. The image forming apparatus according to claim 1, further comprising a toner feeding container that supplies toner to be used in the image carrier unit, wherein the toner feeding container is a container integrated with the waste toner container.
 3. The image forming apparatus according to claim 1, wherein the image carrier unit is, inside the casing, disposed at a position closer to the one side of the first direction than the waste toner container.
 4. The image forming apparatus according to claim 1, wherein the image carrier unit includes an exit port for discharging the waste toner and an exit port shutter that opens and closes the exit port, the waste toner container includes a reception inlet for receiving the waste toner discharged from the exit port and a reception inlet shutter that opens and closes the reception inlet, and the exit port shutter and the reception inlet shutter open the exit port and the reception inlet when the exit port and the reception inlet are disposed at positions where the exit port and the reception inlet face each other.
 5. The image forming apparatus according to claim 4, wherein the image carrier unit further includes a discharge passageway that guides the waste toner to the exit port to the other side of the first direction, the waste toner container further includes a reception passageway that guides the waste toner received from the reception inlet to the other side of the first direction, and the discharge passageway is arranged in an upper position than the reception passageway in a gravity direction.
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